SIMATS ENGINEERING

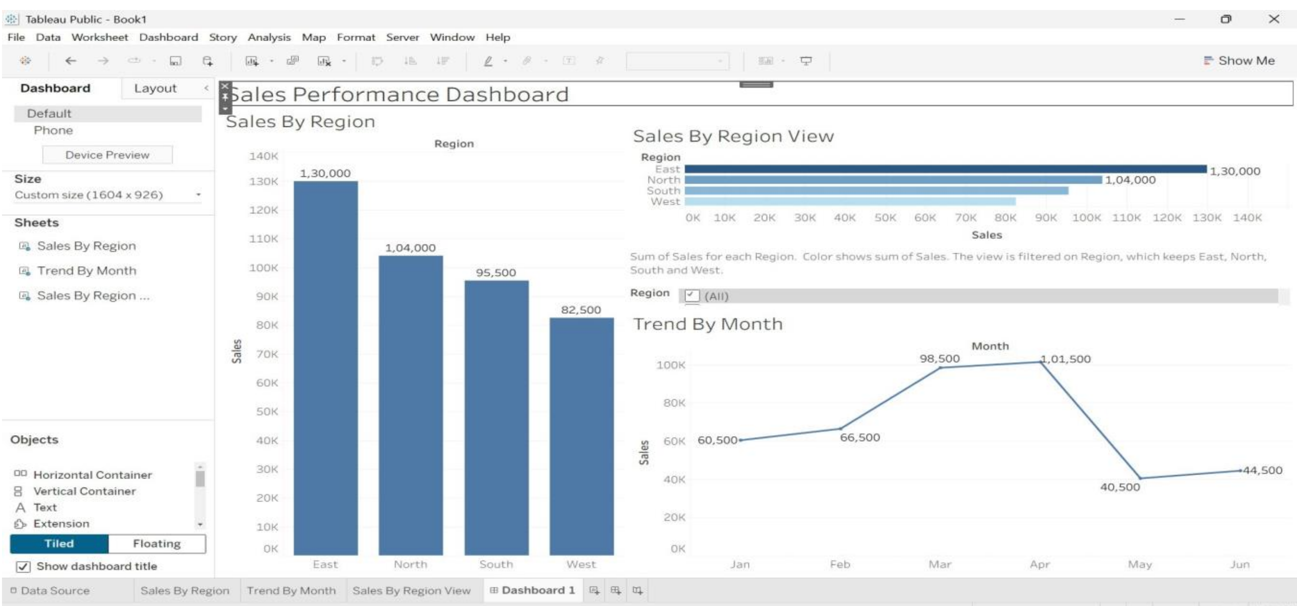
ASSESSMENT TOOL III— Interactive Dashboard Project

Code of Conduct:

I Karthik. C(192573008) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Karthik.C

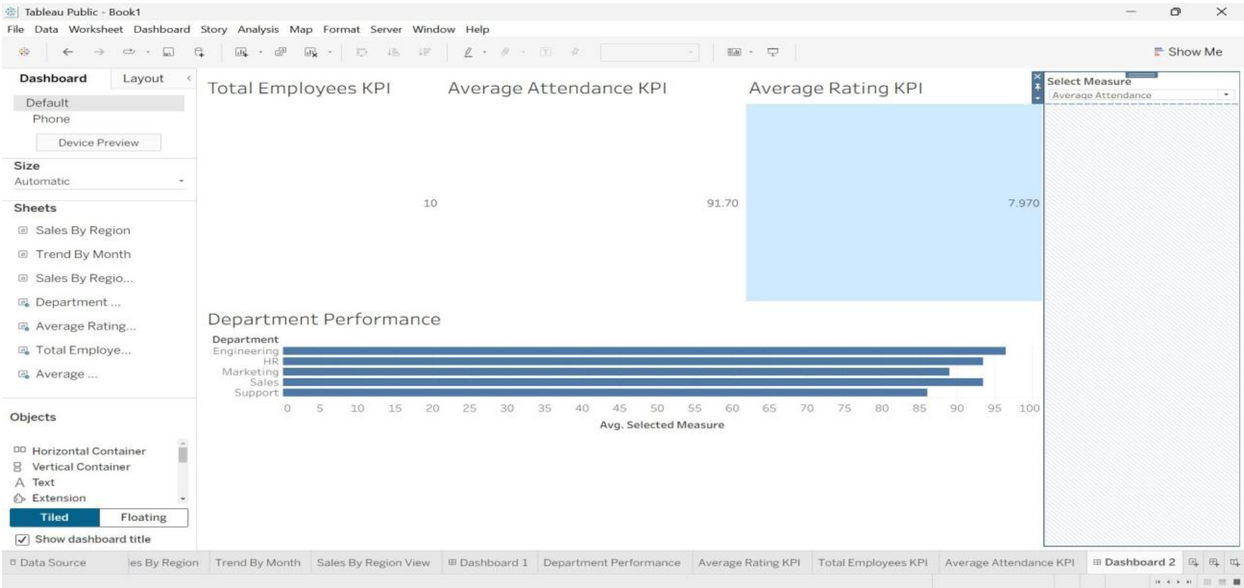
Q1. Building a Sales Performance Dashboard



Interpretation

The Sales Performance Dashboard provides an interactive comparison of regional sales and monthly sales trends. The East region has the highest total sales of 130,000, followed by North with 104,000. West has the lowest sales at 82,500. The monthly trend shows sales increasing from January to April, reaching a peak of 101,500 in April, followed by a sharp decline in May. The Region filter allows users to interactively analyze data for selected regions across all visualizations.

Q2. KPI Dashboard with



Interpretation of the Dashboard

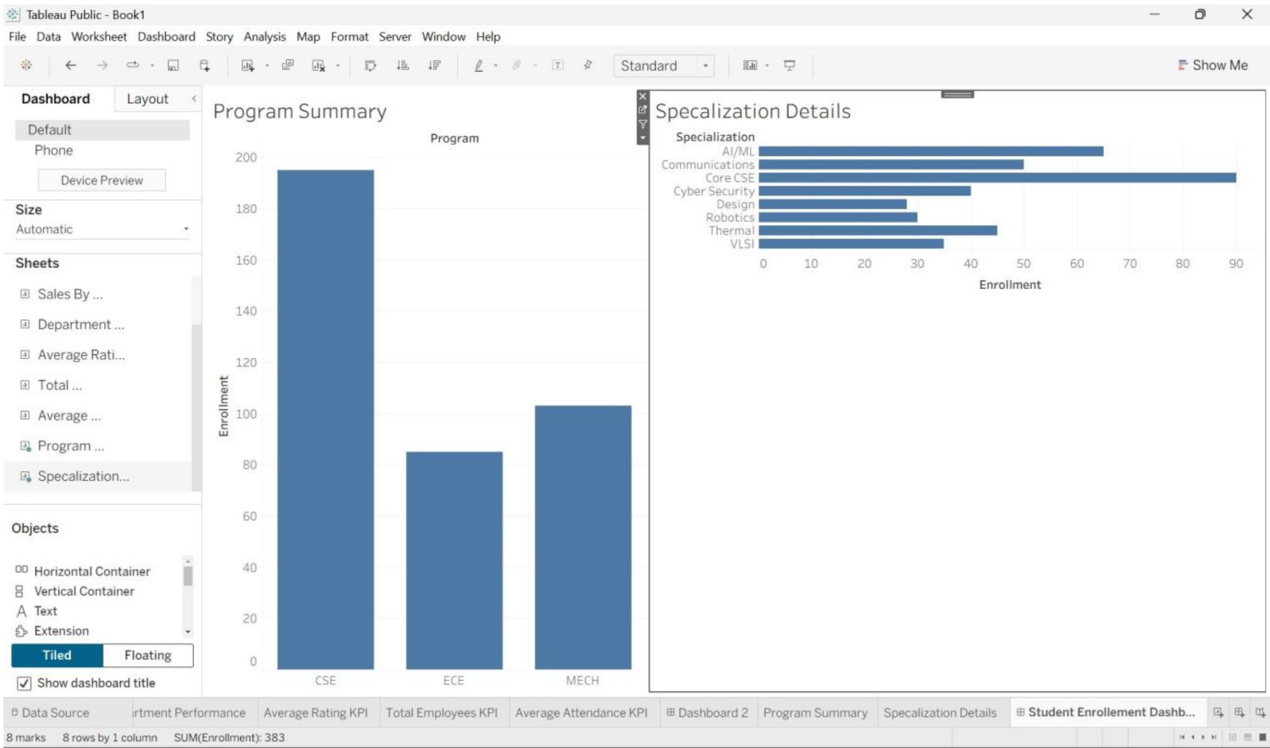
The dashboard presents an overview of employee performance across different departments.

- The organization has a total of 10 employees.
- The average attendance is 91.70%, indicating a generally high level of employee attendance.
- The average performance rating is 7.97 out of 10, showing good overall employee performance.
- In the Department Performance chart, the departments can be compared based on the selected measure.
- When Average Attendance is selected, Engineering has the highest performance, while Support has the lowest value among the departments.
- The Select Measure dropdown makes the dashboard interactive, allowing the user to analyze department performance using different measures such as Attendance and Rating.

Short conclusion

Overall, the dashboard shows that employee attendance and performance are generally good. Engineering performs the best based on the selected measure, while some departments may need improvement. The interactive measure selection helps in comparing departmental performance effectively.

Q3. Drill-Down Dashboard with Dashboard



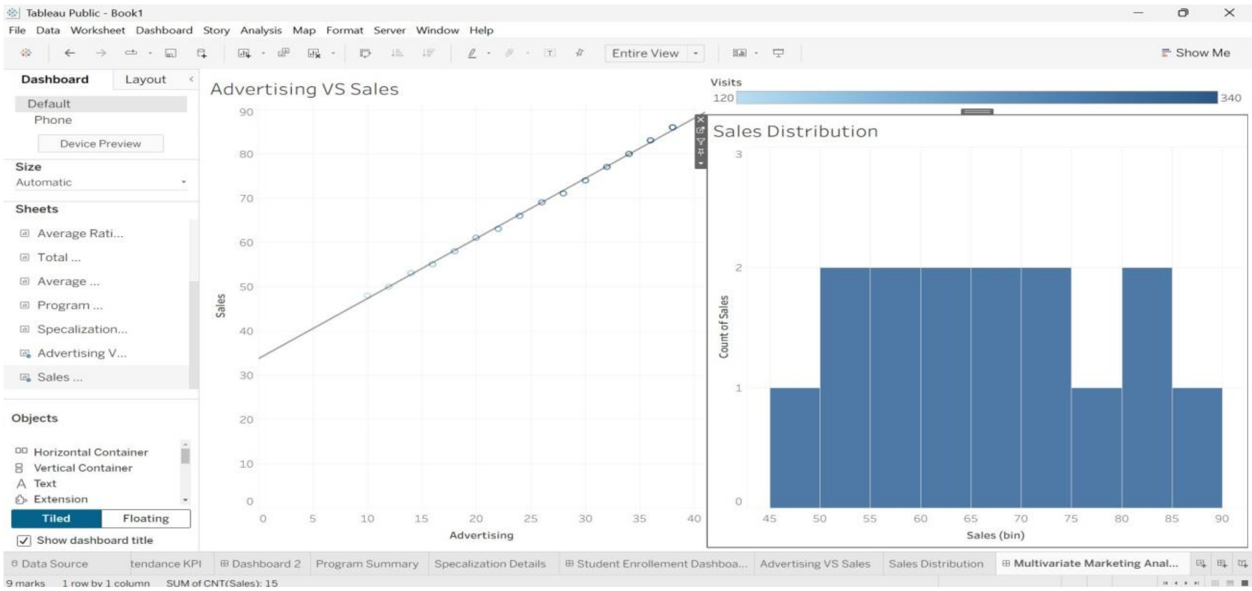
Interpretation

The dashboard provides a drill-down analysis of student enrollment from the Program level to the Specialization level. The Program Summary chart displays the total enrollment for CSE, ECE, and MECH. When a user clicks a Program bar, the Specialization Details chart is filtered to display only the specializations belonging to that selected program. This improves data exploration because users can focus on relevant details instead of viewing all specializations at once. However, this approach depends on user interaction and may require additional actions to compare multiple programs simultaneously.

Limitation

One limitation is that the user can view details for only the selected program at a time, making direct comparison between multiple programs less convenient.

Q4. Multivariate Analysis



Relationship Analysis

The scatterplot shows a strong positive relationship between Advertising expenditure and Sales. As advertising expenditure increases from 10 to 38, sales increase from 48 to 86. The points follow a clear upward pattern, and the trend line has a positive slope, indicating that higher advertising expenditure is associated with higher sales.

Website Visits Analysis

Website Visits also increase along with Advertising and Sales. Using color encoding for Visits provides additional information because it allows the user to observe three variables in a single chart. Higher advertising and sales values generally correspond to higher website visits.

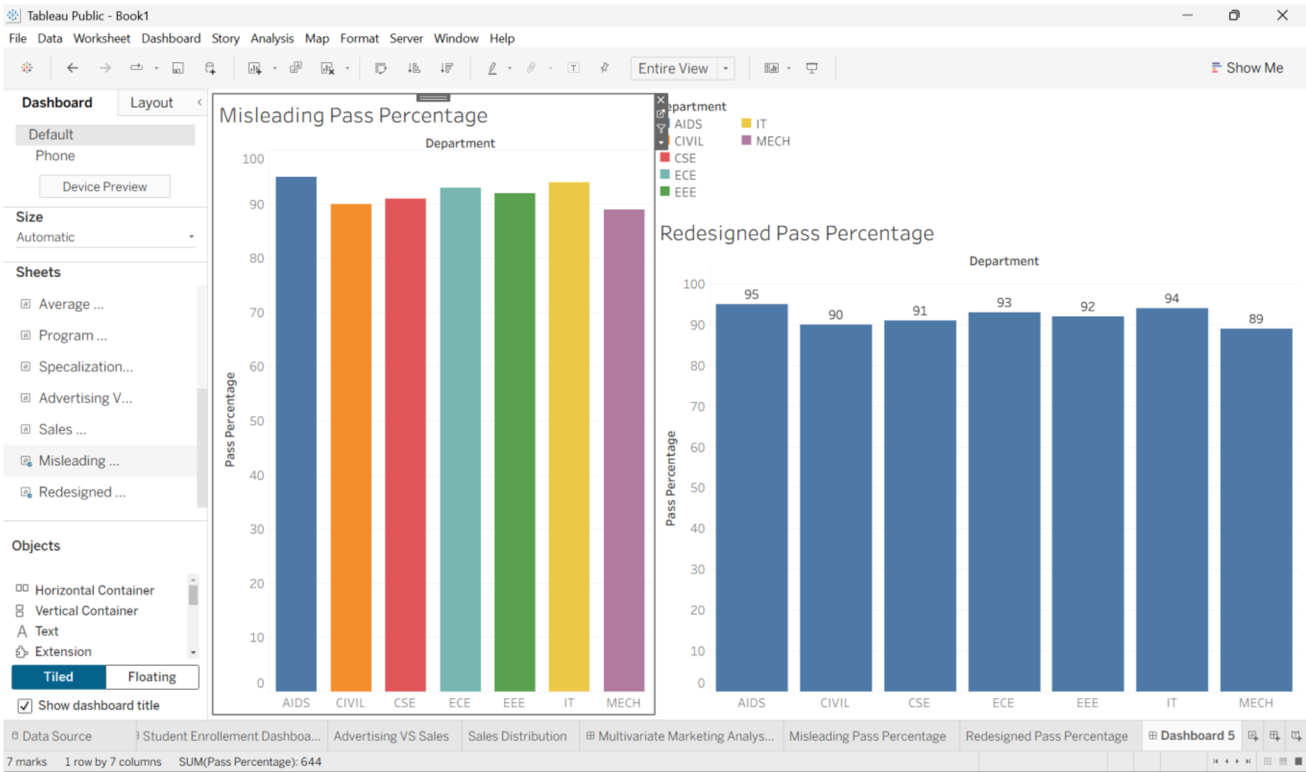
Sales Distribution

The histogram/box plot summarizes the distribution of Sales values. The sales values are spread from approximately 48 to 86, showing a relatively consistent increasing pattern without major outliers.

Visual Clutter Critique

Using color for Website Visits adds useful multivariate information without significantly cluttering the scatterplot. However, using both color and size simultaneously may make the visualization more difficult to interpret. Therefore, using only one additional visual encoding, such as color, is recommended.

Q5. Misleading Dashboard and



Interpretation

The misleading visualization uses a truncated y-axis starting at 85 instead of 0. This exaggerates the visual differences between departments even though the actual pass percentages range only from 89% to 95%, a difference of just 6 percentage points. Using multiple unrelated colors and visually emphasizing one department can further influence the viewer's perception.

The redesigned visualization uses a y-axis starting from 0 and ending at 100, providing an appropriate baseline for comparing percentages. The departments are sorted in descending order, a consistent color scheme is used, and data labels clearly display the exact values. Therefore, the redesigned chart communicates the differences more honestly and makes comparison easier.